EXPERIMENT 7

Design and configure the hub network using cisco packet tracer

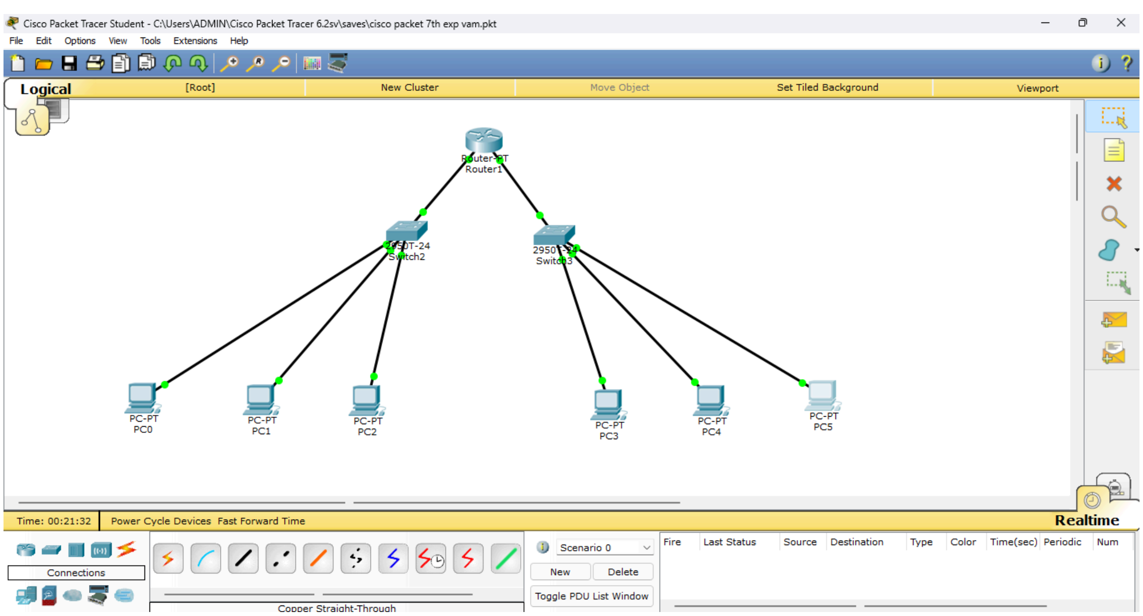
AIM:

To divide the IP block 192.168.10.0/25 into two subnets that can support at least 3 hosts each.
To configure the subnets and verify connectivity between computers using PDU (ping) packets.

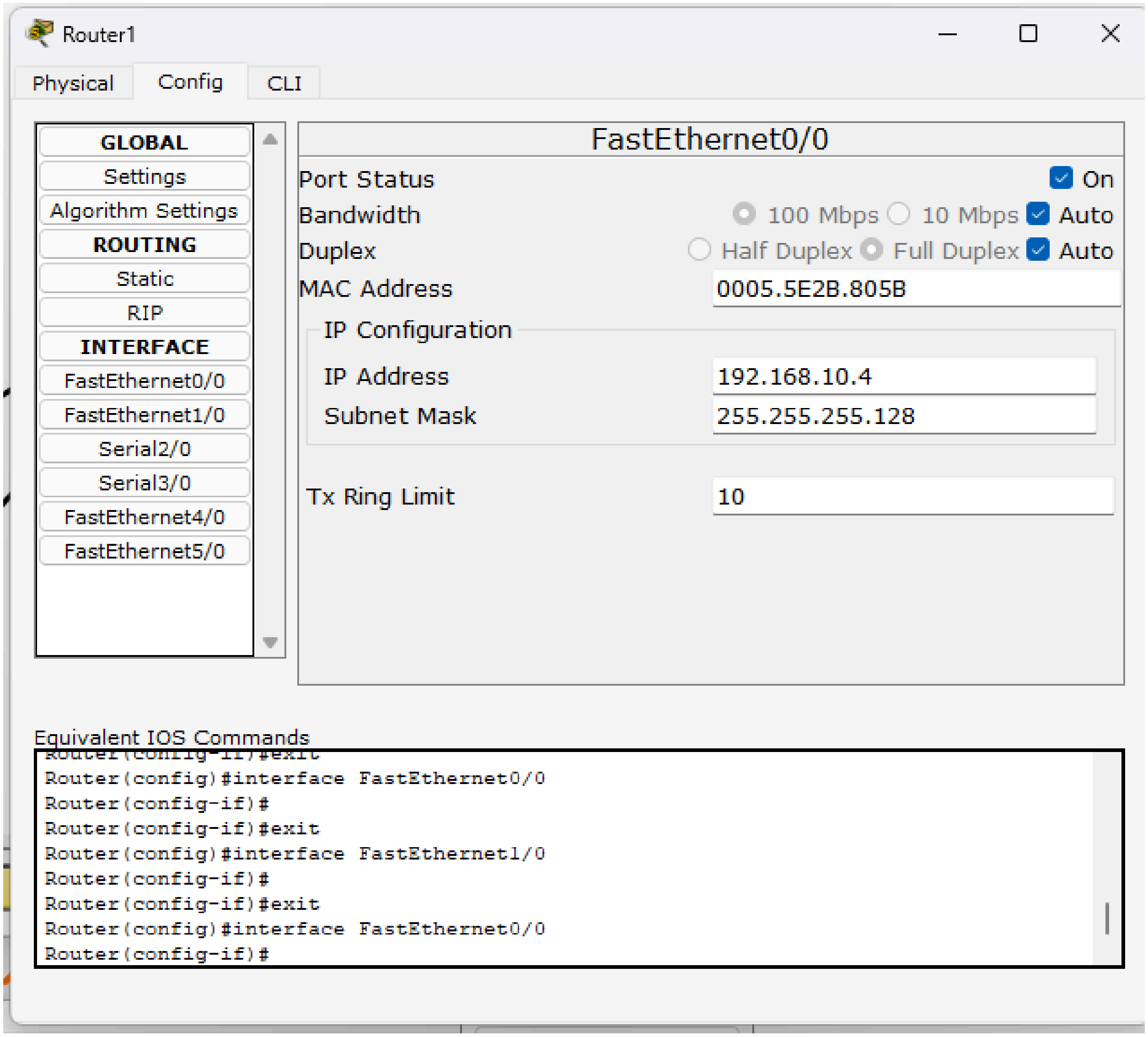
PROCEDURE :

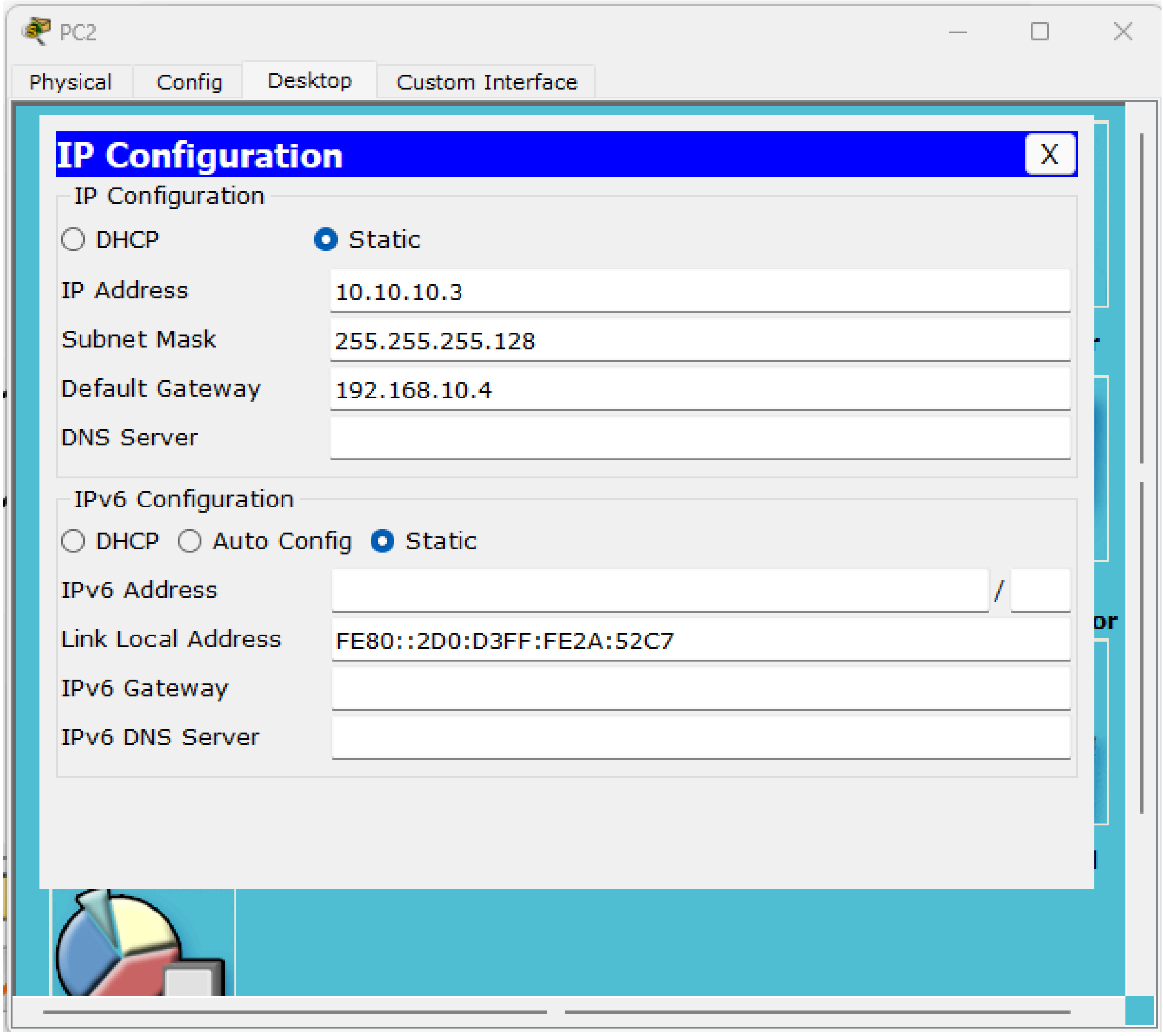
1. Determine subnetting: Split 192.168.10.0/25 into two /26 subnets → Subnet 1: 192.168.10.0/26, Subnet 2: 192.168.10.64/26.
2. Identify host ranges:
3. Subnet 1: 192.168.10.1 – 192.168.10.62 (Broadcast: 192.168.10.63)
4. Subnet 2: 192.168.10.65 – 192.168.10.126 (Broadcast: 192.168.10.127)
5. Assign IP addresses to at least 3 computers in each subnet and configure subnet mask (255.255.255.192).
6. Connect systems using a switch/router (if inter-subnet communication is needed, configure router interfaces with gateway IPs).
7. Test connectivity by sending PDU (ping) packets within and between subnets to verify communication.

BLOCK DIAGRAM :



OUTPUT :


Router1
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0005.5E2B.805B
IP Configuration
IP Address
192.168.10.4
Subnet Mask
255.255.255.128
Tx Ring Limit
10
Equivalent IOS Commands
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#



RESULT :

The network was successfully divided into two subnets with proper IP allocation. Connectivity between computers within and across subnets was verified using PDU packets.